

FEDERAL CIRCUIT REVIEW

DOCTRINE OF EQUIVALENTS

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Despite rumors of its demise when the Supreme Court decided *Festo Corp. v. Shoketsu Konzoku Kogyo Kabushiki Co.*, 535 U.S. 722 (2002), the doctrine of equivalents ("DOE") remains alive and well. Several recent Federal Circuit rulings have affirmed its continued viability and clarified its applicability, scope and limitations.

This issue will discuss a recent opinion where the Federal Circuit considered the applicability of the DOE to claims with numerical-range limitations. We will also look at the court's holdings on proving equivalence of separately patented devices, showing disclaimer of subject matter during prosecution, and limiting the DOE when its application would vitiate a claim limitation. Finally, we will consider why certain patent holders were estopped, or not estopped, from asserting infringement under the DOE.

The DOE prohibits one from avoiding infringement liability by making only "insubstantial changes and substitutions . . . which, though adding nothing, would be enough to take the copied matter outside the [literal scope of the] claim, and hence outside the reach of law." *Graver Tank & Mfg. Co. v. Linde Air Prods. Co.*, 339 U.S. 605, 607 (1950). The "essential inquiry" is whether "the accused product or process contain[s] elements identical or equivalent to each claimed element of the patented invention." *Warner-Jenkinson Co., Inc. v. Hilton Davis Chem. Co.*, 520 U.S. 17, 40 (1997). Equivalence is determined by either (1) the "function-way-result" test or (2) the "insubstantial differences" test. *Warner-Jenkinson*, 520 U.S. at 39. Under the former, the accused product infringes if, on a limitation-by-limitation basis, it "performs substantially the same function in substantially the same way with substantially the same result." *Crown Packaging Tech., Inc. v. Rexam Beverage Can Co.*, 599 F.3d 1308, 1312 (Fed. Cir. 2009). The second test mandates that "[a]n element in the accused device is equivalent to a claim limitation if the only differences between the two are insubstantial." *Honeywell Int'l, Inc. v. Hamilton Sundstrand Corp.*, 370 F.3d 1131, 1139 (Fed. Cir. 2004). Like literal infringement, infringement under the DOE is a question of fact. *Crown Packaging*, 599 F.3d at 1312. However, the DOE is limited by several legal limitations, including prosecution history estoppel, claim vitiation and ensnarement of the prior art. See *Warner-Jenkinson*, 520 U.S. at 39.

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Of the nine cases in which the Federal Circuit substantially addressed equivalents over the past sixteen months, the court (1) vacated three judgments of no infringement under the DOE, (2) affirmed two verdicts of infringement under the DOE, and (3) barred the patentee from asserting the DOE in the four remaining cases, for reasons including prosecution history estoppel and a failure to provide evidence of infringement on a limitation-by-limitation basis.

THE DOE APPLIES TO CLAIMS WITH SPECIFIC NUMERICAL RANGE LIMITATIONS IF THE ACCUSED VALUE IS “INSUBSTANTIALLY DIFFERENT” FROM THE CLAIMED RANGE

In *Adams Respiratory Therapeutics, Inc. v. Perrigo Co.*, 616 F.3d 1283 (Fed. Cir. 2010), the patent concerned extended release formulations of the expectorant guaifenesin. *Id.* at 1285. The claim limitation at issue was the value of AUC¹ of “at least” 3500 hr*ng/mL (nanograms per milliliter per hour). *Id.* at 1291. The accused product had a maximum AUC value of 3493.38 hr*ng/mL, which was 99.8% of the minimum claimed value. *Id.* The defendant argued that, because the claim did not use words of approximation for the range, the patentee could not expand the limitation under the DOE. *Id.* The district court granted summary judgment of no infringement because the “at least” claim limitation was an absolute lower limit and allowing the plaintiff to show infringement under the DOE would vitiate the 3500 hr*ng/mL claim limitation. *Id.* (citing *Quantum Corp. v. Rodime, PLC*, 65 F.3d 1577 (Fed. Cir. 1995)).

The Federal Circuit vacated and remanded, finding that the DOE may apply to claims requiring a specific numeric range without vitiating the numerical range limitation. *Adams Respiratory*, 616 F.3d at 1291. The court explained that “[t]he mere existence of a numerical value or range in a claim, *absent more limiting language in the intrinsic record*, does not preclude application of the DOE.” *Adams Respiratory*, 616 F.3d at 1292 (emphasis added). The real issue in determining infringement of such claims is whether the accused product has a value that is “insubstantially different” from the claimed range. *Id.*

Cases Referenced

Adams Respiratory Therapeutics, Inc. v. Perrigo Co., 616 F.3d 1283 (Fed. Cir. 2010).
Quantum Corp. v. Rodime, PLC, 65 F.3d 1577 (Fed. Cir. 1995).

WHERE THE ALLEGED EQUIVALENT IS SEPARATELY PATENTED, EQUIVALENCE IS NOT TANTAMOUNT TO OBVIOUSNESS

In *Siemens Med. Solutions USA, Inc. v. Saint-Gobain Ceramics & Plastics*, 637 F.3d 1269 (Fed. Cir. 2011), the Federal Circuit ruled that even when an alleged equivalent is claimed in a separate patent, the burden to prove infringement under the DOE is by a preponderance of the evidence.

At issue were two patents related to scintillator crystals used in the medical imaging technique of positron emission tomography (“PET”). Siemens’ ‘080 patent related to scintillator crystals consisting of cerium-doped lutetium oxyorthosilicate (“LSO”). Saint-Gobain manufactured scintillator crystals

¹ AUC refers to the area under a plasma concentration versus time curve, i.e., the total amount of guaifenesin absorbed by the subject. *Id.* at 1291.

consisting of cerium-doped lutetium-yttrium oxyorthosilicate ("LYSO"), and was licensed under a later patent, the '420, which disclosed LYSO scintillator crystals containing from .01% to 99.99% yttrium. *Id.* at 1275–76. The jury found infringement of the '080 patent under the DOE. *Id.* at 1276.

On appeal Saint-Gobain argued that the jury was wrongly instructed to apply the preponderance of evidence standard. Saint-Gobain explained that where an alleged equivalent is separately patented, a heightened evidentiary burden, "clear and convincing evidence," should instead be required because a finding of equivalence "constructively invalidated" the patent covering the alleged equivalent. *Id.* at 1278. Saint Gobain cited *Festo Corp. v. Shoketsu Kinzoku Kogyo Kabushiki Co.*, 493 F.3d 1368, 1379–80 (Fed. Cir. 2007), where the Federal Circuit stated "when a device that incorporates the purported equivalent is in fact the subject of a separate patent, a finding of equivalency, while perhaps not necessarily legally foreclosed, *is at least considerably more difficult to make out*" and "there is a strong argument that an equivalent cannot be both non-obvious and insubstantial." *Siemens*, 637 F.3d at 1278–79 (emphasis added).

The Federal Circuit rejected the argument, however, and affirmed the judgment of infringement. *Id.* at 1274. The court explained that *Festo* concerned whether an equivalent was unforeseeable as of the patent application's filing date" and that the *Festo* language merely indicated that "where, as here, the alleged equivalent is claimed in a separate patent, this fact, when weighed by the fact-finder together with all other relevant evidence, may make equivalency 'considerably more difficult to make out' by a preponderance of the evidence." *Siemens*, 637 F.3d at 1280. Thus, the existence of a patent does not exclude the possibility that the patented device is equivalent to another. Instead, "non-obviousness of the accused device, evidenced by the grant of a United States patent, is relevant to the issue of whether the change therein is substantial." *Id.* at 1281 (citing *Zygo Corp. v. Wyko Corp.*, 79 F.3d 1563, 1570 (Fed. Cir. 1996)).

The court also rejected the argument that a finding of equivalence "constructively invalidated" the '420 patent for four reasons. First, an infringement verdict by a preponderance of the evidence is insufficient to invalidate another patent. *Id.* at 1282. Second, equivalence and obviousness are analyzed under different frameworks: equivalence requires that the difference between the patented and accused invention be "insubstantial," whereas obviousness requires the analysis of predictability in prior art and extrinsic factors such as commercial success. *Id.* Third, equivalency is determined at the time of alleged infringement, while obviousness asks whether the claimed invention would have been obvious when it was made. *Id.* Finally, the extent of infringement found did not equal the scope of the claim: whereas the jury found that only the 10% yttrium crystals infringed the '080 patent, the '420 patent covered the range from .01% to 99.99% yttrium. *Id.*

Judge Prost dissented, postulating that "a separately-patented (and thus presumptively nonobvious) substitution cannot be 'insubstantial' unless some fact distinguishes the equivalence finding from the PTO's earlier nonobviousness determination." *Id.* at 1293. This "fact" could be demonstrated by a "finding that, from the time of the separately-patented invention to the time of accused infringement, the state of the art advanced in such a way as to make a previously non-obvious substitution obvious." *Id.* In Judge Prost's view, "[i]n the absence of such a finding, . . . a separately-patented substitution may not be held equivalent." *Id.*

Cases Referenced

Siemens Med. Solutions USA, Inc. v. Saint-Gobain Ceramics & Plastics, 637 F.3d 1269 (Fed. Cir. 2011).
Festo Corp. v. Shoketsu Kinzoku Kogyo Kabushiki Co., 493 F.3d 1368 (Fed. Cir. 2007).
Zygo Corp. v. Wyko Corp., 79 F.3d 1563 (Fed. Cir. 1996).

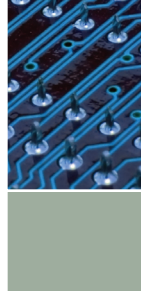
APPLICATION OF THE DOE CANNOT VITIATE A CLAIM LIMITATION

Trading Technologies International, Inc. v. eSpeed, Inc., 595 F.3d 1340 (Fed. Cir. 2010), involved patents for software for displaying the market for commodities traded electronically. The software included a “static display of prices” that eliminated the problem of moving displays causing traders to inadvertently execute trades at unintended prices. *Id.* at 1345–46. Based on the patent claims, specification and prosecution history, the district court construed the term “static display of prices” as “display of prices comprising price levels that do not change positions unless a *manual* re-centering command is received.” *Id.* at 1352 (internal citations omitted). In light of this interpretation, the accused devices, which occasionally re-centered the price columns automatically in response to changes in the market, or had price levels automatically drifting towards the center of the display after every change in the market, did not literally infringe. In addition, the patentee was precluded from asserting the DOE because that would vitiate the claim element. *Id.* at 1354, 1353–55.

The Federal Circuit affirmed, noting that the claims contained a limitation that “the price axis *never* changes positions unless by manual re-centering or re-positioning.” *Id.* at 1354. Further, during prosecution, when the Patent Office rejected the “static display” claims as vague and required the inventors “to claim to what extent, to what degree, and on what basis the displays change,” the inventors explained that “the values in the price column . . . *do not change* (unless a re-centering command is received).” *Id.* (internal citations omitted).

The inventors argued that the claim contained the term “comprising” and thus the preferred embodiment could not be read into the claim, but instead the claim should be interpreted as covering un-recited features such as automatic re-centering. *Id.* at 1354. However, the court pointed out that the inventors expressly stated in the specification that “the present invention” addressed the problem of the moving price axis with “a one-click centering feature,” thus indicating that the re-centering feature is both manual and applicable to the whole invention rather than just an embodiment. *Id.* at 1353–54. In fact, “allowing the price axis to automatically change positions would defy the invention’s goal to ensure fast and accurate execution of trades” and would mean that “the invention would present the same problem as the prior inventions” if the price axis moved automatically, even in rare instances. *Id.* at 1354 (internal citations omitted). In rejecting Trading Technologies’ argument, the court remarked that “‘comprising’ is not a weasel word with which to abrogate claim limitations.” *Id.* at 1353–54 (quoting *Spectrum Int’l v. Sterilite Corp.*, 164 F.3d 1372, 1380 (Fed. Cir. 1998)).

Having agreed with the district court that the claims at issue were limited to manual re-centering during prosecution, the Federal Circuit reiterated that “under prosecution history estoppel, a patentee may not seek to recapture as an equivalent subject matter surrendered during prosecution,” *Id.* at 1355 (citing *Warner-Jenkinson Co., Inc. v. Hilton Davis Chemical Co.*, 520 U.S. 17, 39 n.8 (1997)), and that the all-elements rule bars asserting the DOE where the equivalence would “entirely vitiate a



particular claim element.” *Id.* Because the specification and prosecution history clearly indicated that the claim “forb[ade] all automatic re-centering,” while eSpeed’s accused products featured occasional automatic re-centering or drifting of price levels, the difference was not subtle but instead lay “at the heart of the advantages of the patented invention over prior art.” *Trading Technologies*, 595 F.3d at 1356–57.

Cases Referenced

Trading Technologies International, Inc. v. eSpeed, Inc., 595 F.3d 1340 (Fed. Cir. 2010).

Spectrum Int’l v. Sterilite Corp., 164 F.3d 1372 (Fed. Cir. 1998).

Warner-Jenkinson Co., Inc. v. Hilton Davis Chemical Co., 520 U.S. 17 (1997).

PROSECUTION HISTORY ESTOPPEL LIMITS THE DOE WHEN THE SCOPE OF COVERAGE THAT HAS BEEN SURRENDERED IS CLEAR

1. *INTERVET INC.*

In *Intervet Inc. v. Merial Ltd.*, 617 F.3d 1282 (Fed. Cir. 2010), the Federal Circuit addressed the proper scope of the estoppel entailed by a narrowing claim amendment.

The patent concerned a pathogenic type of the (usually harmless) porcine circovirus that causes disease among livestock pigs, which the patentee named “PCV-2” (porcine circovirus-2). *Id.* at 1284–85. The non-pathogenic, previously known variety is “PCV-1.” *Id.* The patentee isolated five strains of the PCV-2 virus and identified them as representative of the pathogenic variety. *Id.* at 1285. The disclosure stated that the sequenced strains exhibited 96% nucleotide homology² with each other, and only 76% nucleotide homology with PCV-1. *Id.* The disclosure further identified thirteen open reading frames, or “ORFs” (i.e., stretches of gene containing a sequence of nucleotide bases that may encode, or be translated into, a discrete protein), for PCV-2, and stated that nine of the ORFs are unique to PCV-2 and the remaining four are present in both PCV-1 and PCV-2. *Id.*

Intervet, a vaccine maker, sought declaratory judgment that its pig vaccine did not infringe the patent. *Id.* at 1284, 1286. The district court found no literal infringement. *Id.* at 1286–87. The district court then found that the prosecution history narrowed the ORF claim to solely the nine ORFs specific to PCV-2 but not co-occurring in PCV-1 and estopped the patentee from claiming infringement under the DOE. *Id.* at 1291.

The Federal Circuit vacated the judgment and remanded. The court agreed that amendment of the ORF claim was substantially related to patentability under *Festo Corp. v. Shoketsu Konzoku Kogyo Kabushiki Co.*, 535 U.S. 722, 740 (2002). *Id.* at 1291. The Federal Circuit clarified, however, that this amendment estopped only certain arguments under the DOE. Thus, the patentee would be estopped from arguing that ORFs of pathogenic circoviruses found in other organisms, or in PCV-1, are equivalent to the ORFs of PCV-2. But the patentee is not estopped from claiming that a pathogenic porcine viral sequence with over 99% nucleotide homology with one of the five representative strains is equivalent to that strain. *Id.* at 1292.

² Homology refers to the similarity of the DNA/RNA building blocks sequence.

2. FUNAI ELECTRIC

In *Funai Electric Co. v. Daewoo Electronics Corp.*, 616 F.3d 1357 (Fed. Cir. 2010), the Federal Circuit held that prosecution history estoppel did not bar asserting infringement under the DOE where the limitation at issue was not the basis for claim rejection during prosecution.

The patent concerned a mechanism for preventing propagation of driving motor noise and vibration on a tape recorder deck. *Id.* at 1365. The patent claimed “electrical insulation” of various components. *Id.* The jury found that Daewoo’s product was equivalent with regard to the nature of the insulating material. *Id.* at 1368. Daewoo argued, however, that prosecution history estoppel should apply because the claims were rejected for obviousness concerning the presence of an insulating material. *Id.* at 1368–69. The Federal Circuit disagreed. *Id.* at 1369. Quoting *Festo Corp. v. Shoketsu Kinzoku Kogyo Kabushiki Co.*, 535 U.S. 722, 741 (2002), the court explained that in “some cases, . . . the amendment cannot reasonably be viewed as surrendering a particular equivalent”, for example when the “rationale underlying the amendment bear[s] no more than a tangential relation to the equivalent in question.” *Funai*, 616 F.3d at 1369 (internal quotations omitted). The court held that while the presence of an insulating material was rejected for obviousness in certain claims, it was “apparent that the *nature* of the insulating material was not a factor in the allowance” of the remaining claims, as this aspect was “not at issue during prosecution.” *Id.* at 1369. Thus, the district court correctly held that the cancellation of certain claims “did not surrender access to equivalency with respect to the insulating material.” *Id.*

3. LAZARE KAPLAN

In *Lazare Kaplan Int’l, Inc. v. Photoscribe Techs., Inc.*, 628 F.3d 1359 (Fed. Cir. 2010), Lazare sued for infringement of its patent for a system using a laser to create a series of microscopic spots on the surface of gemstones to authenticate and track precious stones. *Id.* at 1364. The patent included a method of “controlling the directing” of laser energy based on marking instructions and imaging. *Id.* The district court construed the limitation to mean “controlling is based on the marking instructions generated by the operator of the machine, and *automatic feedback* derived from optical images of the gemstone *during the laser burn process.*” *Id.* at 1367 (emphases added). Based on this construction, the jury found that Photoscribe did not infringe under the DOE. *Id.* at 1365. Lazare appealed, arguing that the limitation should be construed to permit not only automatic control, but also manual imaging before and during the laser burn process. *Id.* at 1367–68.

The Federal Circuit agreed with Lazare and vacated the verdict of no infringement. The court first noted that the specification contemplated a human operator, stating “[t]he optical feedback system also allows the operator to design an inscription, locate the inscription on the [gem], verify the marking process and archive or store an image of the [gem] and formed markings.” In addition, the specification disclosed that where a single pass of the laser is sufficient to inscribe a gemstone, “an automated optical feedback system may reliably control operation,” but, when multiple passes are necessary, “user control may be desirable [using] a real time image on a computer monitor.” *Id.* at 1369 (internal citations omitted). Thus, the specification disclosed both automated and manual feedback control before and during the laser burn process. *Id.* at 1368–69.

The court also rejected Photoscribe's argument that statements during prosecution show that the patentee considered the "controlling the directing" limitation to require automatic computer control. *Id.* at 1369–70. The court concluded that the prosecution statements fell short of the clear and unmistakable disavowal necessary for the doctrine of prosecution disclaimer to apply. *Id.* at 1370 (citing *Omega Eng'g, Inc. v. Raytek Corp.*, 334 F.3d 1314, 1325–26 (Fed. Cir. 2003)).

Cases Referenced

Intervet Inc. v. Merial Ltd., 617 F.3d 1282 (Fed. Cir. 2010).

Festo Corp. v. Shoketsu Konzoku Kogyo Kabushiki Co., 535 U.S. 722 (2002).

Funai Electric Co. v. Daewoo Electronics Corp., 616 F.3d 1357 (Fed. Cir. 2010).

Lazare Kaplan Int'l, Inc. v. Photoscribe Techs., Inc., 628 F.3d 1359 (Fed. Cir. 2010).

Omega Eng'g, Inc. v. Raytek Corp., 334 F.3d 1314 (Fed. Cir. 2003).

DISTINGUISHING PRIOR ART DURING PROSECUTION CAN RESULT IN DISCLAIMER OF CLAIM SCOPE

Spine Solutions, Inc. v. Medtronic Sofamor Danek USA, Inc., 620 F.3d 1305 (Fed. Cir. 2010), involved a patent for an intervertebral implant, used to replace spinal disks. The patent disclosed that the implant "includes an upper part and a lower part, each of which has a 'support face' for an adjacent vertebra [and] [e]ach of the upper and lower parts has a *single anchor*, or keel, that is centrally positioned on the support face." *Id.* at 1308–09. Medtronic's implant was a two-piece implant arranged as an articulating ball-and-socket joint such that the top and bottom pieces each had an anchoring structure consisting of two tabs separated by a one mm gap. *Id.* at 1315. The district court held that Medtronic's two-tab anchoring structure qualified as a single anchor because the gap between the anchors was a "diagonal slot" not excluded from the construction of "single anchor," and thus Medtronic's implant infringed both literally and under the DOE. *Id.* at 1316.

The Federal Circuit reversed, holding that the accused device plainly had two anchors separated by a gap, while the claim was for one anchor. Therefore, there was no literal infringement. *Id.* at 1317. The court also reversed the grant of summary judgment on the issue of equivalence because the prosecution history estopped the patentee from arguing that two prongs were equivalent to one:

During prosecution, applicants expressly distinguished the claimed invention over [prior art] by asserting that "a reference disclosing two anchors does not disclose a device affirmatively claiming a single anchor . . . reciting a negative limitation in the form of a *single* element is not disclosed by prior art which teaches more than one element." This is a clear disclaimer of claim scope, and prosecution history estoppel therefore bars [plaintiff] from arguing that a two-anchor device is equivalent to the claimed implant.

Id.

Cases Referenced

Spine Solutions, Inc. v. Medtronic Sofamor Danek USA, Inc., 620 F.3d 1305 (Fed. Cir. 2010).

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